

Dietary Protein Modulates Experimental Colitis - Linear Mixed Effects Model

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Linear Mixed Effects Model

Data

Import data CSV

```
df <- read_csv("LMM_database.csv")
```

Filter the database for beef protein diet `Beef_Cellulose` or pea protein diet `Pea_Cellulose` across experiments.

```
df_BP_PP <- df %>% filter(Diet %in% c("Beef_Cellulose", "Pea_Cellulose")) %>% filter(ExpNo  
  ↪ %in% c("DM12", "DM17", "DM19"))
```

Log10 normalize fecal lipocalin-2 fLCN2 data. All fLCN2 values are >0.

```
df_BP_PP$fLCN2 <- log10(df_BP_PP$fLCN2)
```

Build Linear Mixed Effects Models

fLCN2 We fitted a linear mixed effects model with fixed slope, REML=True (best fit estimator) to test the effects of diet (fixed effect, `Beef_Cellulose` vs `Pea_Cellulose`), sex (fixed effect, male vs female), the interaction of diet x sex, experiment batch `ExpNo` (fixed effect), and cage membership (random effect, `UniversalCageNumber`) on fecal lipocalin-2 level fLCN2. Preliminary analysis demonstrated that inclusion of `ExpNo` as a fixed effect term improved model fit.

```
# Preliminary analysis of model with and without ExpNo as fixed term, REML=FALSE.  
BP_PP_LMM_model_no_ExpNo <- lmer(fLCN2 ~ Diet * Sex + (1|UniversalCageNumber), data =  
  ↪ df_BP_PP, REML=FALSE)  
BP_PP_LMM_model_WITH_ExpNo <- lmer(fLCN2 ~ Diet * Sex + ExpNo + (1|UniversalCageNumber),  
  ↪ data = df_BP_PP, REML=FALSE)
```

```
## boundary (singular) fit: see help('isSingular')
```

```

# Compare model fit with and without ExpNo term
anova(BP_PP_LMM_model_no_ExpNo, BP_PP_LMM_model_WITH_ExpNo)

## Data: df_BP_PP
## Models:
## BP_PP_LMM_model_no_ExpNo: fLCN2 ~ Diet * Sex + (1 | UniversalCageNumber)
## BP_PP_LMM_model_WITH_ExpNo: fLCN2 ~ Diet * Sex + ExpNo + (1 | UniversalCageNumber)
##               npar      AIC      BIC logLik -2*log(L)  Chisq Df
## BP_PP_LMM_model_no_ExpNo      6 145.97 158.53 -66.984    133.97
## BP_PP_LMM_model_WITH_ExpNo     8 142.72 159.47 -63.359    126.72 7.2494  2
##               Pr(>Chisq)
## BP_PP_LMM_model_no_ExpNo
## BP_PP_LMM_model_WITH_ExpNo    0.02666 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#Fixed slope, REML=TRUE (best fit estimator)
BP_PP_LMM_model <- lmer(fLCN2 ~ Diet * Sex + ExpNo + (1|UniversalCageNumber), data =
  ↪ df_BP_PP, REML=TRUE)

#LMM results
summary(BP_PP_LMM_model)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: fLCN2 ~ Diet * Sex + ExpNo + (1 | UniversalCageNumber)
## Data: df_BP_PP
##
## REML criterion at convergence: 135.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.96046 -0.67354 -0.00121  0.51944  2.79071
##
## Random effects:
## Groups              Name      Variance Std.Dev.
## UniversalCageNumber (Intercept) 0.01427  0.1195
## Residual                      0.52734  0.7262
## Number of obs: 60, groups: UniversalCageNumber, 18
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)    3.01962    0.26707 13.59848  11.306  2.7e-08 ***
## DietPea_Cellulose -1.31465    0.28911 18.35973  -4.547  0.000238 ***
## SexM             0.07160    0.26409 10.67206   0.271  0.791477
## ExpNoDM17        -0.61141    0.26438 12.58742  -2.313  0.038360 *
## ExpNoDM19        -0.64032    0.25649  7.32719  -2.496  0.039745 *
## DietPea_Cellulose:SexM 0.03924    0.39074 19.73833   0.100  0.921015
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) DtP_C1 SexM  ENDM17 ENDM19

```

```
## DietP_Cl1ls -0.533
## SexM -0.500 0.455
## ExpNoDM17 -0.606 0.072 -0.023
## ExpNoDM19 -0.657 0.131 -0.003 0.618
## DtP_Cl1l:SM 0.357 -0.733 -0.659 -0.013 -0.044
```

```
#Statistical test for fixed effect terms (test if fixed effect term improves models
↪ explanation of the data) - diet, sex, diet x sex interaction, ExpNo
anova(BP_PP_LMM_model)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##          Sum Sq Mean Sq NumDF    DenDF F value    Pr(>F)
## Diet      22.7050  22.7050      1  19.8759 43.0557 2.221e-06 ***
## Sex        0.1100   0.1100      1   6.7861  0.2087  0.66209
## ExpNo      3.7931   1.8965      2   8.2790  3.5964  0.07514 .
## Diet:Sex   0.0053   0.0053      1  19.7383  0.0101  0.92101
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Statistical test for random effect terms (test if random effect term improves models
↪ explanation of the data) - cage membership
ranova(BP_PP_LMM_model)
```

```
## ANOVA-like table for random-effects: Single term deletions
##
## Model:
## fLCN2 ~ Diet + Sex + ExpNo + (1 | UniversalCageNumber) + Diet:Sex
##              npar logLik    AIC    LRT Df Pr(>Chisq)
## <none>              8 -67.639 151.28
## (1 | UniversalCageNumber)  7 -67.654 149.31 0.029177 1 0.8644
```

```
#Extract confidence intervals
confint(BP_PP_LMM_model)
```

```
## Computing profile confidence intervals ...
```

```
## Warning in nextpar(mat, cc, i, delta, lowcut, upcut): Last two rows have
## identical or NA .zeta values: using minstep
```

```
## Warning in FUN(X[[i]], ...): non-monotonic profile for .sig01
```

```
## Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
## .sig01: falling back to linear interpolation
```

```
## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
## collapsing to unique 'x' values
```

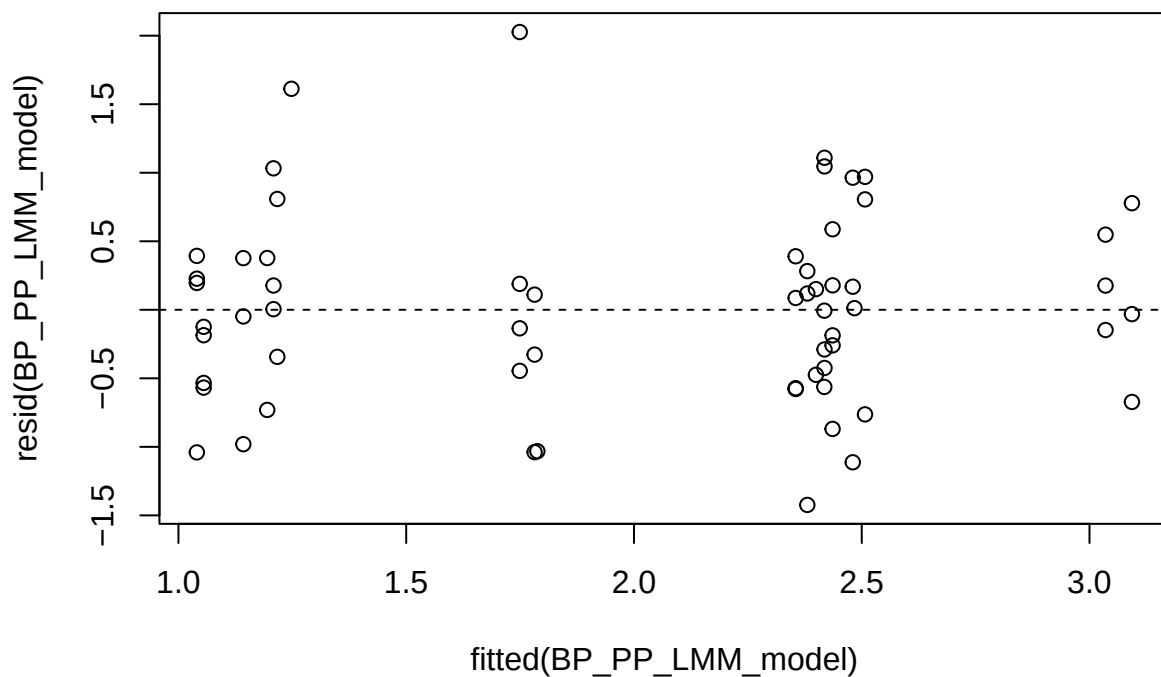
```
##              2.5 %    97.5 %
## .sig01          0.0000000 0.3479624
## .sigma          0.5875439 0.8413964
```

```
## (Intercept)          2.5404886  3.5222608
## DietPea_Cellulose    -1.8570355 -0.7891775
## SexM                 -0.4124445  0.5532352
## ExpNoDM17            -1.1106711 -0.1401569
## ExpNoDM19            -1.1194851 -0.1893667
## DietPea_Cellulose:SexM -0.6804407  0.7647314
```

Results Interpretation for fLCN2 Model Residuals and Confirmation of Assumptions

The model assumptions were confirmed by analysis of the residuals. Plotting residuals vs fitted showed no systematic patterns, with residuals evenly distributed around zero, confirming the assumptions of linearity and homoscedasticity.

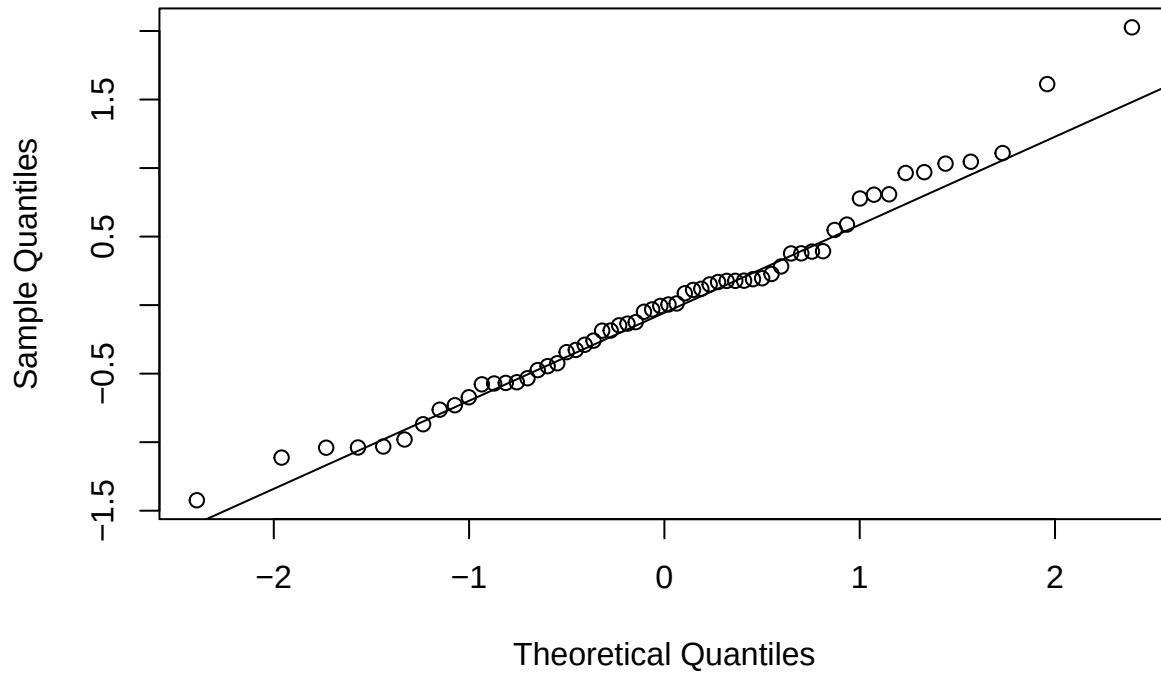
```
plot(resid(BP_PP_LMM_model) ~ fitted(BP_PP_LMM_model))
abline(h = 0, lty = 2)
```



The Q-Q plot of the residuals confirmed approximate normality with points falling on the theoretical diagonal.

```
qqnorm(resid(BP_PP_LMM_model))
qqline(resid(BP_PP_LMM_model))
```

Normal Q-Q Plot



Fixed Effects Analysis - Sex, Diet, Sex x Diet Interaction, ExpNo

Significant contribution of fixed effect terms was assessed by Type III ANOVA. The diet term was highly significant ($F(1,19.9) = 43.1$, $p = 2.221e-06$) with an estimated difference of $-1.31 \log_{10}$ ng/g between *Pea_Cellulose* and *Beef_Cellulose* diets. Sex and interaction of **sex x diet** were non-significant ($F(1,6.8) = 0.21$, $p = 0.66$ and $F(1,19.7) = 0.01$, $p = 0.92$, respectively), demonstrating no sex effects or diet-sex interaction. Experiment batch was retained as a fixed effect term based on preliminary likelihood ratio testing ($p=0.027$); however, the Type III F-test for *ExpNo* was marginal ($p=0.075$), indicating inter-experiment variability was present but had minimal contribution to measured effects.

```
#Statistical test for fixed effect terms (test if fixed effect term improves models
→ explanation of the data) - diet, sex, diet x sex interaction, ExpNo
anova(BP_PP_LMM_model)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq Mean Sq NumDF   DenDF F value    Pr(>F)
## Diet      22.7050  22.7050     1  19.8759  43.0557 2.221e-06 ***
## Sex        0.1100   0.1100     1   6.7861   0.2087  0.66209
## ExpNo      3.7931   1.8965     2   8.2790   3.5964  0.07514 .
## Diet:Sex   0.0053   0.0053     1  19.7383   0.0101  0.92101
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Random Effects Analysis - Cage Membership

Contribution of the random effect term was assessed with likelihood ratio testing by `ranova()`. The random intercept for cage membership *UniversalCageNumber* was not significant (LRT = 0.029, $p = 0.864$), indicat-

ing negligible cage effect. The cage term was retained in the model due to nested cage UniversalCageNumber within experiment ExpNo.

```
#Statistical test for random effect terms (test if random effect term improves models
→ explanation of the data) - cage membership
ranova(BP_PP_LMM_model)
```

```
## ANOVA-like table for random-effects: Single term deletions
##
## Model:
## fLCN2 ~ Diet + Sex + ExpNo + (1 | UniversalCageNumber) + Diet:Sex
##               npar logLik   AIC     LRT Df Pr(>Chisq)
## <none>              8 -67.639 151.28
## (1 | UniversalCageNumber)  7 -67.654 149.31 0.029177 1      0.8644
```

Visualization of sex meta-data in diet groups

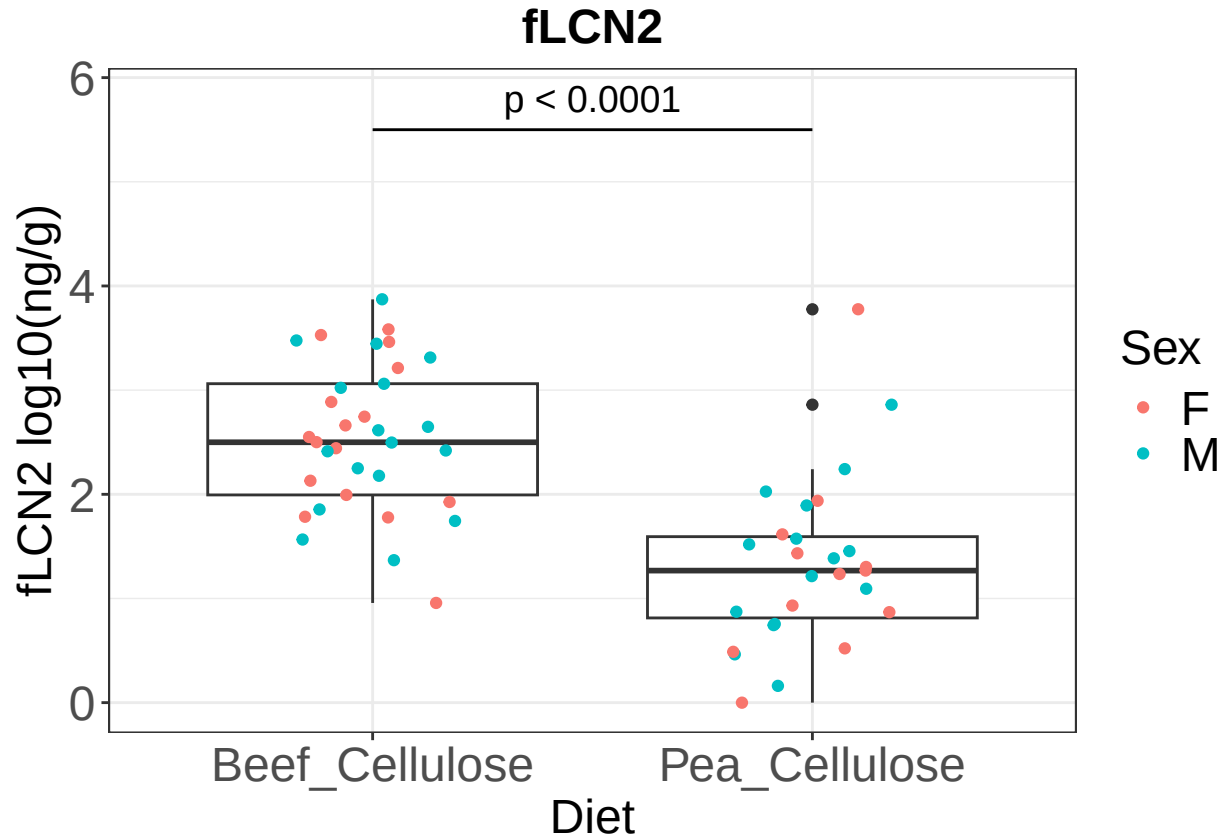
To visually demonstrate how sex distributed within diet groups, sex meta-data was colored on a plot of diet by fLCN2 level, visually confirming that results do not cluster by sex.

```
#Visualization by meta-data
p <- ggplot(df_BP_PP, aes(x=Diet, y=fLCN2)) +
  geom_boxplot() +
  ## plot all data points, colored by meta-data
  geom_jitter(aes(color = Sex),
    position = position_jitter(width=0.2),
    alpha=1) +
  ## Significance bar for Diet comparison
  annotate("segment", x = 1, xend = 2, y = 5.5, yend = 5.5) +
  annotate("text", x = 1.5, y = 5.8, label = "p < 0.0001", size = 5) +
  ##
  labs(title="fLCN2", y="fLCN2 log10(ng/g)") +
  ## Plot theme formatting
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size=18),
    axis.text = element_text(size=18),
    axis.title=element_text(size=18),
    legend.text = element_text(size=18),
    legend.title = element_text(size=18)
  )

print(p)
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Total Histology Score We fitted a linear mixed effects model with fixed slope, REML=True (best fit estimator) to test the effects of diet (fixed effect, **Beef_Cellulose** vs **Pea_Cellulose**), sex (fixed effect, male vs female), the interaction of diet x sex, experiment batch **ExpNo** (fixed effect), and cage membership (random effect, **UniversalCageNumber**) on histology **TotalHistoScore**. Preliminary analysis demonstrated that inclusion of **ExpNo** as a fixed effect term did not improve model fit, so **ExpNo** was dropped from the model.

```
# Preliminary analysis of model with and without ExpNo as fixed term, REML=FALSE.
BP_PP_LMM_model_no_ExpNo <- lmer(TotalHistoScore ~ Diet * Sex + (1|UniversalCageNumber),
  ↪ data = df_BP_PP, REML=FALSE)
BP_PP_LMM_model_WITH_ExpNo <- lmer(TotalHistoScore ~ Diet * Sex + ExpNo +
  ↪ (1|UniversalCageNumber), data = df_BP_PP, REML=FALSE)
# Compare model fit with and without ExpNo term
anova(BP_PP_LMM_model_no_ExpNo, BP_PP_LMM_model_WITH_ExpNo)
```

```
## Data: df_BP_PP
## Models:
## BP_PP_LMM_model_no_ExpNo: TotalHistoScore ~ Diet * Sex + (1 | UniversalCageNumber)
## BP_PP_LMM_model_WITH_ExpNo: TotalHistoScore ~ Diet * Sex + ExpNo + (1 |
  ↪ UniversalCageNumber)
##               npar      AIC      BIC   logLik -2*log(L)  Chisq Df
## BP_PP_LMM_model_no_ExpNo      6 273.66 286.42 -130.83   261.66
## BP_PP_LMM_model_WITH_ExpNo     8 273.22 290.24 -128.61   257.22 4.4394  2
##               Pr(>Chisq)
## BP_PP_LMM_model_no_ExpNo
```

```
## BP_PP_LMM_model_WITH_ExpNo      0.1086
```

```
#Fixed slope, REML=TRUE (best fit estimator)
```

```
BP_PP_LMM_model <- lmer(TotalHistoScore ~ Diet * Sex + (1|UniversalCageNumber), data =  
  ↪ df_BP_PP, REML=TRUE)
```

```
#LMM results
```

```
summary(BP_PP_LMM_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]
```

```
## Formula: TotalHistoScore ~ Diet * Sex + (1 | UniversalCageNumber)
```

```
## Data: df_BP_PP
```

```
##
```

```
## REML criterion at convergence: 258
```

```
##
```

```
## Scaled residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -2.1754 -0.4758 -0.1244  0.5908  3.4283
```

```
##
```

```
## Random effects:
```

```
## Groups              Name          Variance Std.Dev.
```

```
## UniversalCageNumber (Intercept) 1.143    1.069
```

```
## Residual                      3.461    1.860
```

```
## Number of obs: 62, groups: UniversalCageNumber, 18
```

```
##
```

```
## Fixed effects:
```

```
##              Estimate Std. Error    df t value Pr(>|t|)
```

```
## (Intercept)      6.1539    0.6435 14.8480   9.563 9.80e-08 ***
```

```
## DietPea_Cellulose -4.9465    0.9029 34.8088  -5.479 3.82e-06 ***
```

```
## SexM              1.2045    0.8978 15.0800   1.342  0.20
```

```
## DietPea_Cellulose:SexM -1.2594    1.2229 34.3969  -1.030  0.31
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Correlation of Fixed Effects:
```

```
##              (Intr) DtP_Cl SexM
```

```
## DietP_Cl1ls -0.540
```

```
## SexM         -0.717  0.387
```

```
## DtP_Cl1l:SM  0.399 -0.738 -0.596
```

```
#Fixed effects test
```

```
anova(BP_PP_LMM_model)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
```

```
##              Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
```

```
## Diet         287.848  287.848     1 34.397 83.1668 1.05e-10 ***
```

```
## Sex           2.177    2.177     1 10.968  0.6291  0.4445
```

```
## Diet:Sex      3.671    3.671     1 34.397  1.0606  0.3103
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Test if random effects improve fit
ranova(BP_PP_LMM_model)
```

```
## ANOVA-like table for random-effects: Single term deletions
##
## Model:
## TotalHistoScore ~ Diet + Sex + (1 | UniversalCageNumber) + Diet:Sex
##               npar logLik   AIC   LRT Df Pr(>Chisq)
## <none>                6 -129.02 270.04
## (1 | UniversalCageNumber)  5 -130.48 270.95 2.9136  1    0.08783 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Extract confidence intervals
confint(BP_PP_LMM_model)
```

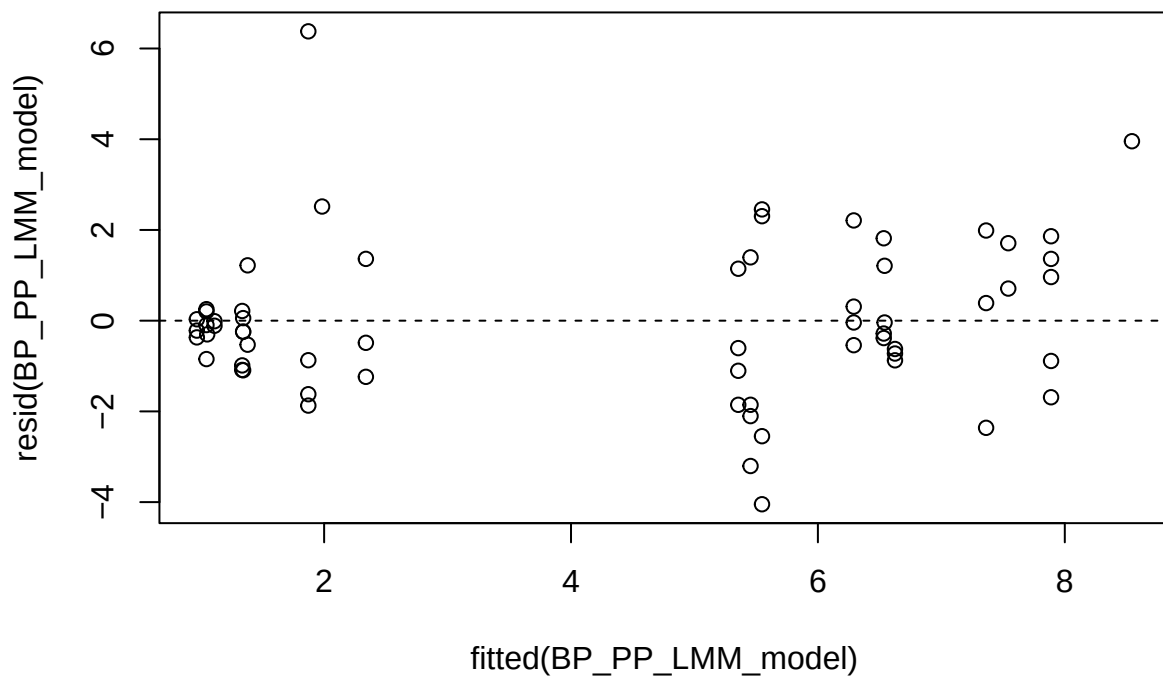
```
## Computing profile confidence intervals ...

##               2.5 %    97.5 %
## .sig01          0.0000000  1.732223
## .sigma          1.5166061  2.334237
## (Intercept)      4.9450492  7.362523
## DietPea_Cellulose -6.6300480 -3.141969
## SexM             -0.4768435  2.891499
## DietPea_Cellulose:SexM -3.5275145  1.098908
```

Results Interpretation for Total Histology Score Model Residuals and Confirmation of Assumptions

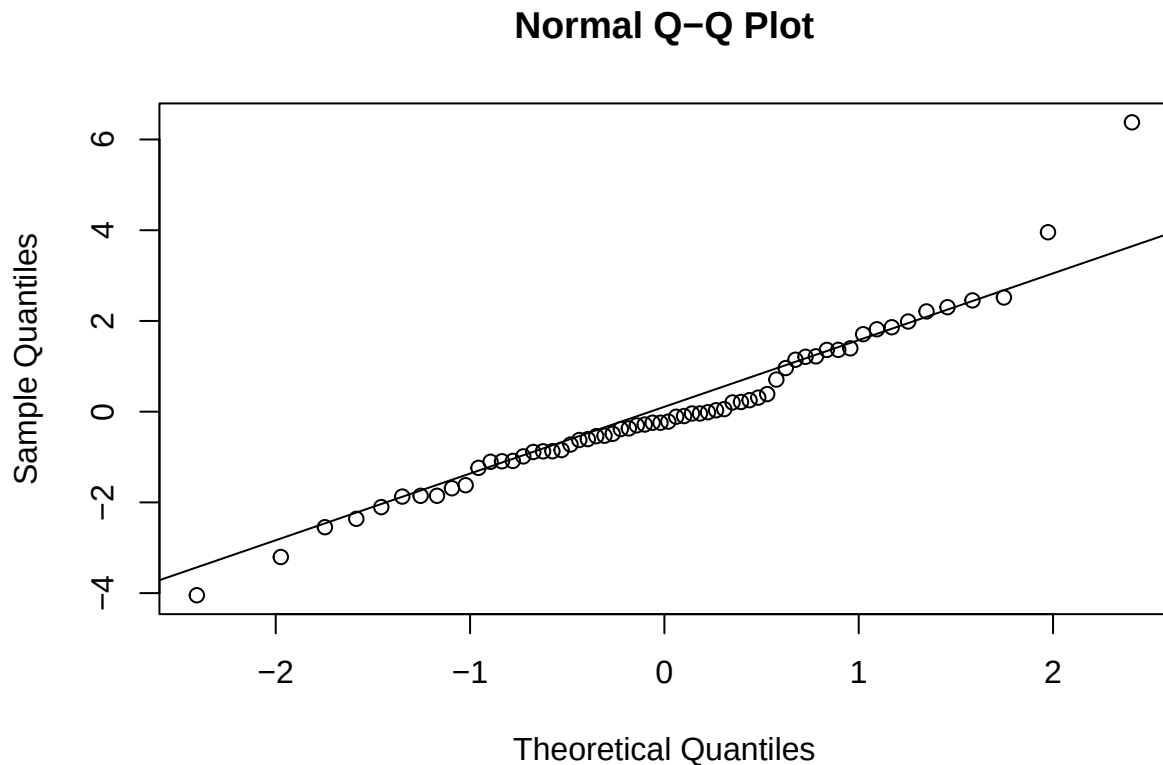
The model assumptions were confirmed by analysis of the residuals. Plotting residuals vs fitted showed no systematic patterns, with residuals evenly distributed around zero, confirming the assumptions of linearity and homoscedasticity.

```
plot(resid(BP_PP_LMM_model) ~ fitted(BP_PP_LMM_model))
abline(h = 0, lty = 2)
```



The Q-Q plot of the residuals confirmed approximate normality with points falling on the theoretical diagonal.

```
qqnorm(resid(BP_PP_LMM_model))  
qqline(resid(BP_PP_LMM_model))
```



Fixed Effects Analysis - Sex, Diet, Sex x Diet Interaction

Significant contribution of fixed effect terms was assessed by Type III ANOVA. The diet term was highly significant ($F(1,34.4) = 83.1$, $p = 1.05e-10$) with an estimated difference of -4.95 points on the 0–20 scale between Pea_Cellulose and Beef_Cellulose diets. Sex and interaction of **sex x diet** were non-significant ($F(1,10.97) = 0.629$, $p = 0.44$ and $F(1,34.4) = 1.06$, $p = 0.31$, respectively), demonstrating no sex effects or diet-sex interaction.

```
#Statistical test for fixed effect terms (test if fixed effect term improves models
→ explanation of the data) - diet, sex, diet x sex interaction, ExpNo
anova(BP_PP_LMM_model)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## Diet      287.848  287.848     1  34.397  83.1668 1.05e-10 ***
## Sex         2.177    2.177     1  10.968   0.6291  0.4445
## Diet:Sex    3.671    3.671     1  34.397   1.0606  0.3103
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Random Effects Analysis - Cage Membership

Contribution of the random effect term was assessed with likelihood ratio testing by `ranova()`. The random intercept for cage membership **UniversalCageNumber** was not significant (LRT = 2.914, $p = 0.088$). The cage term was retained in the model to account for non-independence of cage-mates.

```
#Statistical test for random effect terms (test if random effect term improves models
→ explanation of the data) - cage membership
ranova(BP_PP_LMM_model)
```

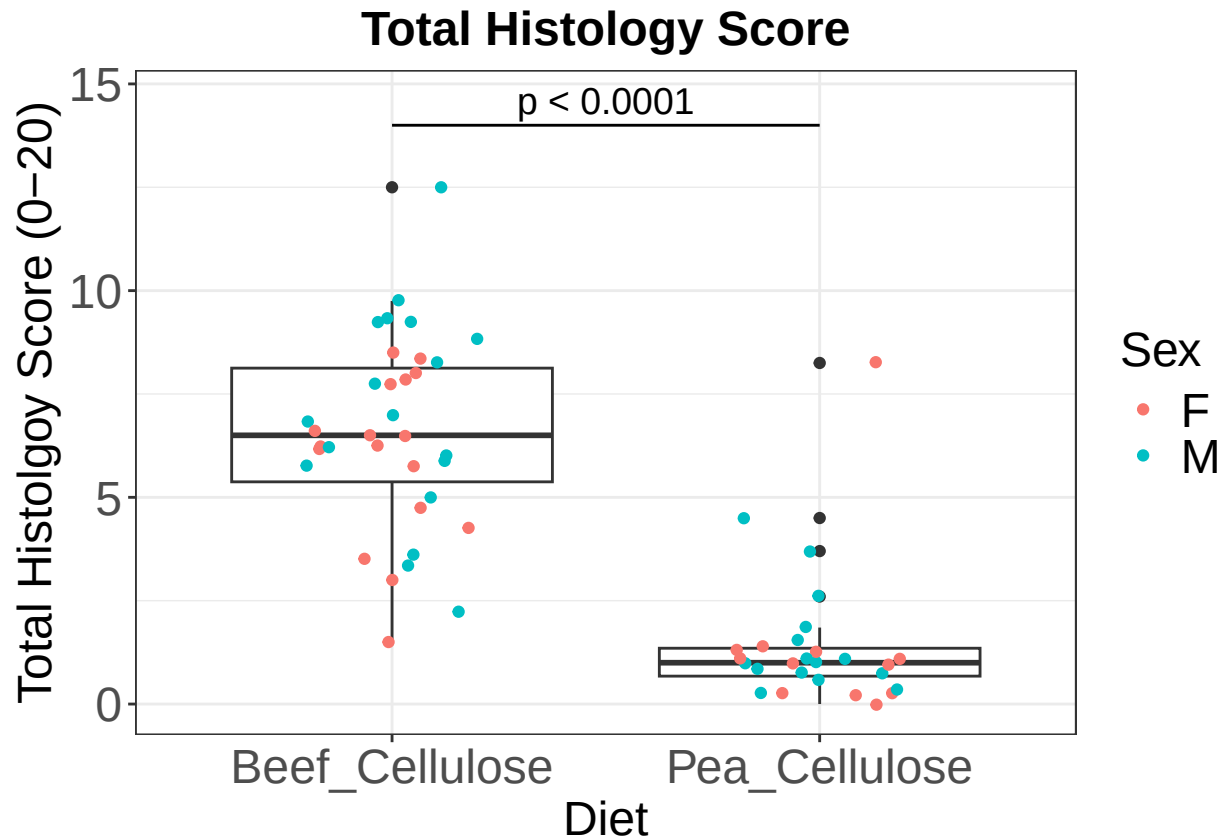
```
## ANOVA-like table for random-effects: Single term deletions
##
## Model:
## TotalHistoScore ~ Diet + Sex + (1 | UniversalCageNumber) + Diet:Sex
##               npar  logLik    AIC    LRT Df Pr(>Chisq)
## <none>                6 -129.02 270.04
## (1 | UniversalCageNumber)  5 -130.48 270.95 2.9136  1    0.08783 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Visualization of sex meta-data in diet groups

To visually demonstrate how sex distributed within diet groups, sex meta-data was colored on a plot of diet by total histology score, visually confirming that results do not cluster by sex.

```
#Visualization by meta-data
p <- ggplot(df_BP_PP, aes(x=Diet, y=TotalHistoScore)) +
  geom_boxplot() +
  ## plot all data points, colored by meta-data
  geom_jitter(aes(color = Sex),
              position = position_jitter(width=0.2),
              alpha=1) +
  ## Significance bar for Diet comparison
  annotate("segment", x = 1, xend = 2, y = 14, yend = 14) +
  annotate("text", x = 1.5, y = 14.6, label = "p < 0.0001", size = 5) +
  ##
  labs(title="Total Histology Score", y="Total Histology Score (0-20)") +
  ## Plot theme formatting
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size=18),
    axis.text = element_text(size=18),
    axis.title=element_text(size=18),
    legend.text = element_text(size=18),
    legend.title = element_text(size=18)
  )

print(p)
```



Total Weight Change We fitted a linear mixed effects model with fixed slope, REML=True (best fit estimator) to test the effects of diet (fixed effect, **Beef_Cellulose** vs **Pea_Cellulose**), sex (fixed effect, male vs female), the interaction of diet x sex, experiment batch **ExpNo** (fixed effect), and cage membership (random effect, **UniversalCageNumber**) on % Weight Change **WeightChange**. Preliminary analysis demonstrated that inclusion of **ExpNo** as a fixed effect term improved model fit.

```
# Preliminary analysis of model with and without ExpNo as fixed term, REML=FALSE.
BP_PP_LMM_model_no_ExpNo <- lmer(WeightChange ~ Diet * Sex + (1|UniversalCageNumber),
  ↪ data = df_BP_PP, REML=FALSE)
BP_PP_LMM_model_WITH_ExpNo <- lmer(WeightChange ~ Diet * Sex + ExpNo +
  ↪ (1|UniversalCageNumber), data = df_BP_PP, REML=FALSE)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Compare model fit with and without ExpNo term
anova(BP_PP_LMM_model_no_ExpNo, BP_PP_LMM_model_WITH_ExpNo)
```

```
## Data: df_BP_PP
## Models:
## BP_PP_LMM_model_no_ExpNo: WeightChange ~ Diet * Sex + (1 | UniversalCageNumber)
## BP_PP_LMM_model_WITH_ExpNo: WeightChange ~ Diet * Sex + ExpNo + (1 |
  ↪ UniversalCageNumber)
##
```

	npair	AIC	BIC	logLik	-2*log(L)	Chisq	Df
--	-------	-----	-----	--------	-----------	-------	----

```

## BP_PP_LMM_model_no_ExpNo      6 407.23 419.90 -197.62      395.23
## BP_PP_LMM_model_WITH_ExpNo    8 397.37 414.26 -190.69      381.37 13.857  2
##                               Pr(>Chisq)
## BP_PP_LMM_model_no_ExpNo
## BP_PP_LMM_model_WITH_ExpNo  0.0009797 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#Fixed slope, REML=TRUE (best fit estimator)
BP_PP_LMM_model <- lmer(WeightChange ~ Diet * Sex + ExpNo + (1|UniversalCageNumber), data
  ↪ = df_BP_PP, REML=TRUE)

## boundary (singular) fit: see help('isSingular')

#LMM results
summary(BP_PP_LMM_model)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: WeightChange ~ Diet * Sex + ExpNo + (1 | UniversalCageNumber)
## Data: df_BP_PP
##
## REML criterion at convergence: 365.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.48063 -0.55707  0.07981  0.56508  2.08342
##
## Random effects:
## Groups              Name             Variance Std.Dev.
## UniversalCageNumber (Intercept)  0.00      0.000
## Residual                      33.71      5.806
## Number of obs: 61, groups: UniversalCageNumber, 18
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)    -5.7443     1.9163 55.0000  -2.998  0.00408 **
## DietPea_Cellulose  11.0813     2.2001 55.0000   5.037 5.43e-06 ***
## SexM             -5.9432     1.9940 55.0000  -2.981  0.00428 **
## ExpNoDM17         0.7907     1.9504 55.0000   0.405  0.68675
## ExpNoDM19        -6.0821     1.8681 55.0000  -3.256  0.00194 **
## DietPea_Cellulose:SexM  8.7448     3.0097 55.0000   2.906  0.00527 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) DtP_Cl SexM  ENDM17 ENDM19
## DietP_Cl1ls -0.524
## SexM        -0.521  0.455
## ExpNoDM17   -0.567  0.033 -0.025
## ExpNoDM19   -0.633  0.096  0.022  0.583
## DtP_Cl1l:SM  0.371 -0.730 -0.663 -0.007 -0.056
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

```

```
#Fixed effects test
anova(BP_PP_LMM_model)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq Mean Sq NumDF DenDF  F value    Pr(>F)
## Diet       3543.0   3543.0     1    55 105.1134 2.267e-14 ***
## Sex         36.8    36.8     1    55   1.0921 0.3005767
## ExpNo       627.6   313.8     2    55   9.3104 0.0003292 ***
## Diet:Sex    284.6   284.6     1    55   8.4422 0.0052724 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Test if random effects improve fit
ranova(BP_PP_LMM_model)
```

```
## ANOVA-like table for random-effects: Single term deletions
##
## Model:
## WeightChange ~ Diet + Sex + ExpNo + (1 | UniversalCageNumber) + Diet:Sex
##           npar  logLik    AIC LRT Df Pr(>Chisq)
## <none>              8 -182.64 381.28
## (1 | UniversalCageNumber)  7 -182.64 379.28  0 1          1
```

```
#Extract confidence intervals
confint(BP_PP_LMM_model)
```

```
## Computing profile confidence intervals ...
```

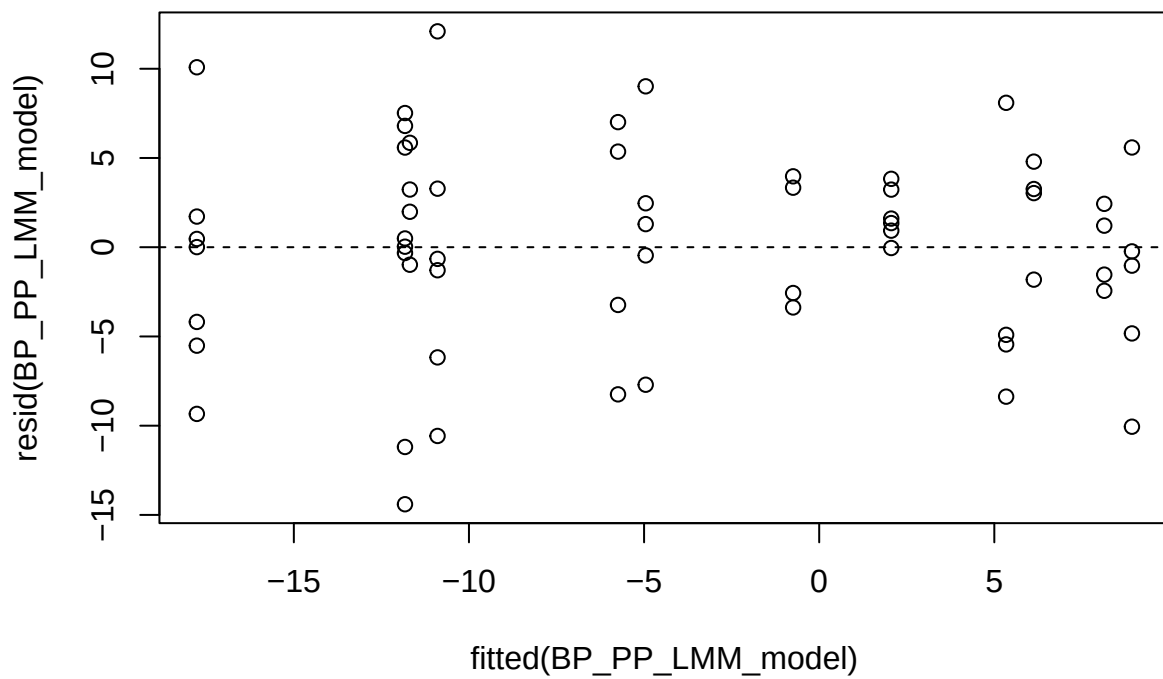
```
## Warning in optwrap(optimizer, devfun, x@theta, lower = x@lower, calc.derivs =
## TRUE, : convergence code 3 from bobyqa: bobyqa -- a trust region step failed to
## reduce q
```

```
##           2.5 %    97.5 %
## .sig01      0.000000  2.223644
## .sigma      4.662400  6.657003
## (Intercept) -9.368748 -2.121119
## DietPea_Cellulose  6.921506 15.238566
## SexM          -9.714272 -2.172912
## ExpNoDM17     -2.897168  4.478326
## ExpNoDM19     -9.613780 -2.554549
## DietPea_Cellulose:SexM  3.054243 14.435401
```

Results Interpretation for Weight Change Model Residuals and Confirmation of Assumptions

The model assumptions were confirmed by analysis of the residuals. Plotting residuals vs fitted showed no systematic patterns, with residuals evenly distributed around zero, confirming the assumptions of linearity and homoscedasticity.

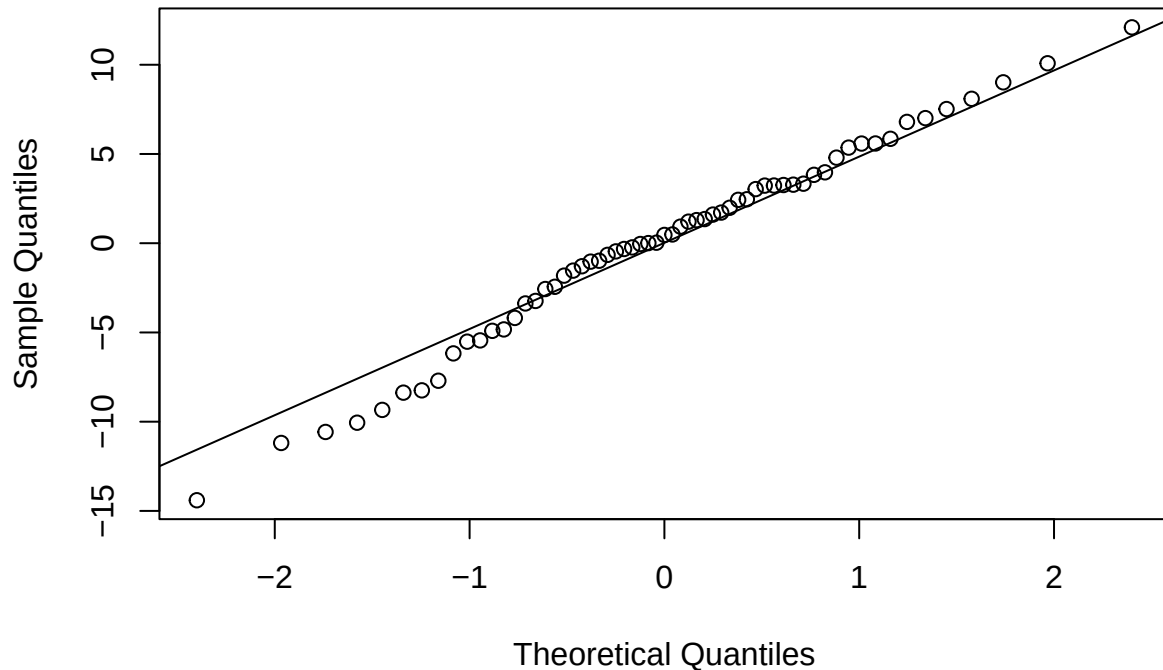
```
plot(resid(BP_PP_LMM_model) ~ fitted(BP_PP_LMM_model))
abline(h = 0, lty = 2)
```



The Q-Q plot of the residuals confirmed approximate normality with points falling on the theoretical diagonal.

```
qqnorm(resid(BP_PP_LMM_model))  
qqline(resid(BP_PP_LMM_model))
```

Normal Q-Q Plot



Fixed Effects Analysis - Sex, Diet, Sex x Diet Interaction, ExpNo

Significant contribution of fixed effect terms was assessed by Type III ANOVA. The diet term was highly significant ($F(1,55) = 105.11$, $p = 2.267e-14$) with an estimated difference of 11.08 % between **Pea_Cellulose** and **Beef_Cellulose** diets. Sex was not-significant ($F(1, 55) = 1.09$, $p = 0.30$), but the interaction of **sex x diet** was significant ($F(1,55) = 8.44$, $p = 0.005$), indicating diet-sex interaction for % weight change. Experiment batch was retained as a fixed effect term based on preliminary likelihood ratio testing ($p=0.001$). The Type III F-test for **ExpNo** was significant ($p=0.0003$), indicating inter-experiment variability was present for % weight change.

```
#Statistical test for fixed effect terms (test if fixed effect term improves models
→ explanation of the data) - diet, sex, diet x sex interaction, ExpNo
anova(BP_PP_LMM_model)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##          Sum Sq Mean Sq NumDF DenDF  F value    Pr(>F)
## Diet      3543.0  3543.0     1    55 105.1134 2.267e-14 ***
## Sex         36.8   36.8     1    55  1.0921 0.3005767
## ExpNo       627.6  313.8     2    55  9.3104 0.0003292 ***
## Diet:Sex    284.6   284.6     1    55  8.4422 0.0052724 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Random Effects Analysis - Cage Membership

Contribution of the random effect term was assessed with likelihood ratio testing by **ranova()**. The random intercept for cage membership **UniversalCageNumber** was not significant ($LRT = 0$, $p = 1$), indicating no

cage effect. The cage term was retained in the model due to nested cage `UniversalCageNumber` within experiment `ExpNo`.

```
#Statistical test for random effect terms (test if random effect term improves models
→ explanation of the data) - cage membership
ranova(BP_PP_LMM_model)
```

```
## ANOVA-like table for random-effects: Single term deletions
##
## Model:
## WeightChange ~ Diet + Sex + ExpNo + (1 | UniversalCageNumber) + Diet:Sex
##               npar logLik   AIC LRT Df Pr(>Chisq)
## <none>              8 -182.64 381.28
## (1 | UniversalCageNumber)  7 -182.64 379.28  0  1          1
```

Post-hoc analysis of Diet x Sex interaction

The significant interaction of `sex x diet` ($F(1,55) = 8.44$, $p = 0.005$) was interrogated using estimated marginal means with the `emmeans()` function. Because experiment number `ExpNo` was a fixed effects term in the LMM, `emmeans()` computes the model-predicted mean for every group after averaging over co-variables, including `ExpNo`.

Diet effect within sex

Post-hoc pairwise analysis demonstrated a significant diet effect in **both** females (estimate = -11.1 % weight change, $p = 0.0001$) **and** males (estimate = -19.8 % weight change, $p < 0.0001$). Males had a larger magnitude weight change in response to diet.

```
#Calculate diet effect within sex
emm_diet_in_sex <- emmeans(BP_PP_LMM_model, ~ Diet | Sex)
summary(emm_diet_in_sex)
```

```
## Sex = F:
## Diet      emmean  SE    df lower.CL upper.CL
## Beef_Cellulose -7.51 1.47 12.70 -10.692   -4.32
## Pea_Cellulose   3.57 1.78  9.04  -0.448    7.59
##
## Sex = M:
## Diet      emmean  SE    df lower.CL upper.CL
## Beef_Cellulose -13.45 1.45 13.74 -16.567  -10.34
## Pea_Cellulose   6.37 1.55 23.33   3.176    9.57
##
## Results are averaged over the levels of: ExpNo
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
#Compute difference between diets within each sex [ie 1) diet effect in females; 2) diet
→ effect in males]
contrast_emm_diet_in_sex <- contrast(emm_diet_in_sex, method = "pairwise")
summary(contrast_emm_diet_in_sex)
```

```
## Sex = F:
## contrast              estimate    SE    df t.ratio p.value
```

```
## Beef_Cellulose - Pea_Cellulose    -11.1 2.32 18.8  -4.773  0.0001
##
## Sex = M:
## contrast          estimate    SE   df t.ratio p.value
## Beef_Cellulose - Pea_Cellulose    -19.8 2.13 25.8  -9.290  <.0001
##
## Results are averaged over the levels of: ExpNo
## Degrees-of-freedom method: kenward-roger
```

Sex effect within diet

Post-hoc pairwise analysis of sex effects within each diet demonstrated males lost significantly more weight than females fed Beef_Cellulose diet (estimate = 5.94 % weight change, $p = 0.0129$), but no significant sex difference was found in males and females fed Pea_Cellulose diet (estimate = -2.80 % weight change, $p = 0.26$). Together, these analyses demonstrate that Beef_Cellulose diet induces significant weight loss in **both** males and females vs Pea_Cellulose diet, but that males have a larger magnitude weight loss, which is driven by sex x diet interaction with Beef_Cellulose diet only, not Pea_Cellulose diet.

```
#Calculate sex effect within diet
emm_sex_in_diet <- emmeans(BP_PP_LMM_model, ~ Sex | Diet)
summary(emm_sex_in_diet)
```

```
## Diet = Beef_Cellulose:
## Sex emmean    SE    df lower.CL upper.CL
## F    -7.51 1.47 12.70  -10.692   -4.32
## M   -13.45 1.45 13.74  -16.567  -10.34
##
## Diet = Pea_Cellulose:
## Sex emmean    SE    df lower.CL upper.CL
## F     3.57 1.78  9.04   -0.448    7.59
## M     6.37 1.55 23.33    3.176    9.57
##
## Results are averaged over the levels of: ExpNo
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
#Compute difference between sexes within each diet [ie 1) diet effect in females; 2) diet
↪ effect in males]
contrast_emm_sex_in_diet <- contrast(emm_sex_in_diet, method = "pairwise")
summary(contrast_emm_sex_in_diet)
```

```
## Diet = Beef_Cellulose:
## contrast estimate    SE    df t.ratio p.value
## F - M          5.94 2.06 12.7    2.889  0.0129
##
## Diet = Pea_Cellulose:
## contrast estimate    SE    df t.ratio p.value
## F - M         -2.80 2.38 13.0   -1.179  0.2595
##
## Results are averaged over the levels of: ExpNo
## Degrees-of-freedom method: kenward-roger
```

Visualization of sex meta-data in diet groups

To visually demonstrate how sex distributed within diet groups, sex meta-data was colored on a plot of diet by % Weight change level.

```
#Visualization by meta-data
p <- ggplot(df_BP_PP, aes(x=Diet, y=WeightChange, shape=Sex)) +
  geom_boxplot() +
  ## plot all data points, colored by meta-data
  geom_jitter(aes(color = Sex),
              position = position_jitterdodge(jitter.width=0.2),
              alpha=1) +
  ## Add significant bars for sex comparisons within diet
  annotate("segment", x = 0.625, xend = 1.375, y = 12, yend = 12) +
  annotate("text", x = 1, y = 15, label = "p = 0.013", size = 5) +
  ## Pea_Cellulose: F vs M, p = 0.26 (ns)
  annotate("segment", x = 1.625, xend = 2.375, y = 22, yend = 22) +
  annotate("text", x = 2, y = 25, label = "ns", size = 5) +

  ##
  labs(title="% Weight Change", y="Weight Change (%)") +
  ## Plot theme formatting
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size=18),
    axis.text = element_text(size=18),
    axis.title=element_text(size=18),
    legend.text = element_text(size=18),
    legend.title = element_text(size=18)
  )

print(p)
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

